# Answer key: Lesson 8, Make a Table

**For the tutor and grown-ups.** Keep this page away from the child until they have finished and checked their own work. Ask *how do you know?* before saying right or wrong.

## Start here (pages 1 and 2)

**S1.** Squares in a row are made with sticks. Look at the picture. Count the sticks.

**Answer: 4; 7; 10**

The first square has 4 sticks. The second square shares one side, so it adds 3 more sticks: 7 in all. The third square adds 3 more: 10 in all.

**Watch for:** Writing 8 for 2 squares (4 + 4): the squares share a side.; Writing 12 for 3 squares.

**S2.** For every 3 red beads there are 2 blue beads. Fill in the table.

**Answer: 9 and 8; 15**

Red beads go up by 3 each time: 3, 6, 9, 12. Blue beads go up by 2 each time: 2, 4, 6, 8. 10 blue beads is 5 groups, and 5 groups have 15 red beads.

**Watch for:** Adding 3 to the blue beads, or 2 to the red beads.; Writing 10 or 12 for the red beads.

**S3.** Siti keeps adding squares to the row. Fill in the table.

**Answer: 13, 16, 19; 3; 22**

The sticks go up by 3 each time, because each new square shares a side: 4, 7, 10, 13, 16, 19. 7 squares: 19 and 3 more is 22 sticks.

**Watch for:** Adding 4 each time.; Stopping at 19 for 7 squares (that is 6 squares).

**S4.** Ravi reads 5 pages each day. The table shows the pages he has read in all. Fill in the table.

**Answer: 15, 25, 30; day 5**

Each day adds 5 pages: 5, 10, 15, 20, 25, 30. 25 pages in all is day 5.

**Watch for:** Writing 5 again in each box.; Adding 4 or 6 instead of 5.

**S5.** Look at the squares in a row. Tick True or False for each statement.

**Answer: true; false; true**

Each new square shares a side, so it adds 3 sticks: True. 4 squares need 13 sticks, not 16: False. 6 squares need 19 sticks: True.

**Watch for:** Saying 4 squares need 4 times 4 = 16 sticks.; Not checking the statement against the table.

## Think harder (page 3)

**T1.** Mi makes triangles in a row with sticks. Fill in the table. Then finish the rule.

**Answer: 21 and 41; add 1**

From 4 to 10 triangles there are 6 more triangles, and each adds 2 sticks: 9 and 12 make 21. Or double 10 is 20, and 1 more is 21. For 20 triangles: double 20 is 40, and 1 more is 41. The rule is: double the number of triangles, then add 1.

**Watch for:** Writing 30 or 60 (3 sticks for every triangle).; Writing 11 for the 10th picture (counting on only one step).

**T2.** Look at rows of triangles made with sticks.

**Answer: No; odd; Yes**

Each row has double the triangles and then 1 more, so the number of sticks is an even number plus 1, which is odd. 20 is even, so no row of triangles uses exactly 20 sticks. 25 is odd: 25 take away 1 is 24, and half of 24 is 12, so 12 triangles use exactly 25 sticks.

**Watch for:** Saying Yes after finding a close number such as 21.; Thinking that every number of sticks is possible.

**T3.** Ben made a table for squares in a row. One number is wrong.

**Answer: 4; 13; 3**

The numbers go up by 3 each time: 4, 7, 10, then 13, 16, 19. The number for 4 squares is 12, but it should be 13.

**Watch for:** Choosing 5 because 16 comes after the wrong number.; Writing 11 or 14.

## Go further (page 4)

**G1.** Picture 100. Look at the tables for rows of triangles and rows of squares.

**Answer: 201; 301; 151**

Triangles: double 100 is 200, and 1 more is 201. Squares: the first square has 4 sticks and each of the other 99 squares adds 3: 99 times 3 is 297, and 4 more is 301. Or 100 times 3 is 300, and 1 more is 301. 50 squares: 50 times 3 is 150, and 1 more is 151.

**Watch for:** Writing 300 or 400 (3 or 4 sticks for every picture).; Writing 200 for the triangles and forgetting the 1.

**G2.** Work back from the number of sticks.

**Answer: 15; 12**

Triangles: 31 minus 1 is 30, and half of 30 is 15. Squares: 37 minus 1 is 36, and 36 shared into groups of 3 is 12.

**Watch for:** Dividing 31 by 3 for the triangles.; Forgetting to take away the 1 first.

**G3.** A row of squares and a row of triangles have the same number of pictures.

**Answer: 10; 20; 5**

10 squares need 31 sticks and 10 triangles need 21 sticks: 10 more. 20 squares need 61 sticks and 20 triangles need 41 sticks: 20 more. 5 squares need 16 sticks and 5 triangles need 11 sticks: 5 more. Each new picture needs 1 more stick for a square than for a triangle.

**Watch for:** Comparing the first pictures only (4 and 3).; Subtracting 10 and 20 from each other.

## Investigation (page 5): Strips left over

**First: 7 strips. The most triangles is 6 (13 strips, 1 left over). The most squares is 4 (13 strips, 1 left over).**
**No: a row of triangles uses an odd number of strips, so 14 never works, and the squares use 4, 7, 10, 13 or 16 strips, so 14 does not work either.**

**Look for:** A child who builds with the strips and counts; a child who uses the table to find how far 14 goes; a child who says why 14 never fits (odd numbers for triangles; 13 then 16 for squares). Credit any clear reason. Do not mark a child down for stopping at one shape.

## Exit ticket (page 5)

**E1.** Each tray holds 7 cups. 1 tray has 7 cups. 2 trays have 14 cups.

**Answer: 21**

Each tray adds 7 cups: 7, 14, 21. 3 trays have 21 cups.

**Watch for:** Writing 20 or 28.; Writing 3 (the number of trays).

**E2.** Mei reads 3 pages each day. How many pages does she read in 5 days? Choose the correct answer. Write its number in the bracket.

**Answer: (3) 15**

Each day adds 3 pages: 3, 6, 9, 12, 15. In 5 days she reads 15 pages.

**Watch for:** Choosing (1) 8, which adds 3 and 5.; Choosing (4) 35.

**E3.** Can exactly 12 sticks make a row of triangles with none left over? Circle Yes or No.

**Answer: No**

A row of triangles uses an odd number of sticks: 3, 5, 7, 9, 11, 13. 12 is even, so it is not possible.

**Watch for:** Saying Yes because 12 is near 11 and 13.; Counting 3 sticks for each triangle: 4 triangles.

## Sneak-peek puzzles (not in the workbook)

1. 12 pencils
2. 14 dots. Each picture has 3 more dots than the one before: picture 4 has 11 and picture 5 has 14.